AshAway - Task 5: Testing Report

Software Engineering for CSAI 2026-2027 | Group 9 | Client: WanderGuard | Hammza Abas, Amin El Amrani, Rayan Fikri, Calvin Kakembo, Mikail Kaya, David Vasilescu | github.com/Davidtzuke/AshAway

Approach and why these tests

AshAway's worst failure is false reassurance: a traveller told nothing is wrong when we simply don't know. We chose tests around the ways that could happen: bad requests, wrong units, future data leaking into the model, and failing feeds. Unit tests check one function with hand-computed answers. Integration tests check that components work together, using real data. **pytest** runs them all automatically, and **GitHub Actions** runs them on every push.

Unit tests: one function, no files or network

ID	Unit	Test input	Expected output (all matched)	Why this test	Passed
U1	validate_request	'atlantis'; end before start; 31 Oct to 1 Nov; end 15 days ahead. Also Rhodes, 2-4 July, today 1 July.	Rejected with unsupported_place; reversed_dates; out_of_season; beyond_horizon. Accepted with 3 days.	Guessing outside our scope would give false advice (FR1, FR9).	14/14
U2	risk_level, cut-offs 0.2 and 0.5	0.199; 0.2; 0.499; 0.5; also 1.01 and NaN.	low; medium; medium; high; ValueError.	Off-by-one errors live at boundaries (FR2).	10/10
U3	is_stale, check_evidence	Value 2 days old; 3 days old; none. Low forecast with 3 fires nearby.	Current; stale; stale. Conflict flagged.	Old data must never look reassuring (FR5).	11/11
U4	parse_weather	30 C, 40%, 5 mm. Rain null. {"error": true}.	303.15 K, 0.40, 0.005 m. NaN, not 0. ValueError.	Wrong units or an invented 0 silently corrupt a score (NFR5).	5/5
U5	window_features, daily_feature_rows	8 days with 2 mm rain 2 days ago. Weather including the day itself. Gap in history.	rain_7d 0.0025, days_since_rain 2. Uses only the day before. No row.	Inputs must match training and not see the future (NFR9).	6/6
U6	lower_risk_alternatives	High risk; candidates at high, medium, low, and low with poor coverage.	Low, then medium. Others excluded.	An equal-risk 'alternative' would mislead (FR6, AI6).	3/3
U7	choose_threshold_f2, fit_calibration, high_threshold	Toy labels and scores with a known answer.	Threshold 0.4; monotonic calibration; lowest score reaching the precision target.	Cut-offs decide what travellers see (NFR9, AI5).	3/3
U8	in_greece	Athens, Rhodes, Corfu. Izmir, Tirana, open sea.	True. False.	Greek results must not include neighbours (AI4).	6/6

Integration tests: several components together

ID	Components	Test input	Expected output (all matched)	Why this test	Passed
I1	Loader + split + data report	114 real Track A samples.	Inputs stop the day before the label; no outcome columns; 36 per split at 2:1; report shows 114 raw, 6 removed, 108 clean.	Guards the fire-day leak we found in the first loader (AI2, NFR2, NFR9).	4/4
I2	Training + calibration + save + load	Same sample.	Reloaded model gives identical predictions; record lists 5 models and all NFR8 metrics.	Each prediction must be reproducible (NFR4, NFR8).	5/5
I3	Feeds + features + model + service	Rhodes, 18-24 July 2023, recorded feeds. Chania, 1-3 June 2023, with no fires, then 1 fire nearby.	High, 7 days, every item labelled, dated and sourced, nothing after 17 July. Low. Insufficient evidence with a conflict warning.	A known real event checks the whole answer (FR2-4, FR8).	6/6
I4	Feeds + service, under failure	Weather HTTP 500. Fire feed down with a low, then a high forecast. Air quality down.	No score or reasons, source unavailable. Insufficient evidence; high stays with a warning. No air item.	Failure must never read as 'safe' (FR5, NFR5).	8/8
I5	Service across destinations	Rhodes, 1-3 July 2023. Rhodes and Chania on the same dates.	1-3 alternatives, strictly lower and sorted. One comparable row per place.	Advice must compare like with like (FR6, FR7).	4/4
I6	Command line + service + feeds	assess rhodes 2023-07-18 2023-07-24. assess atlantis ...; December dates.	Exit 0, risk in words, reasons, labels, disclaimer. Exit 2, reason, no risk level.	This is what a traveller really sees (UR1-4, NFR6).	8/8

Test data

Hand-computed values for unit tests. **A real Mesogeos Track A sample**: 114 samples, 3,420 rows, with real gaps and fire days, small enough for automated runs. **Recorded Open-Meteo responses** from 2023, including the July Rhodes heatwave. **Simulated HTTP 500 errors**, since real services don't fail on demand. **Hand-written FIRMS files** in NASA's documented format, which are not real detections, because our API key isn't set up yet.

Results

All 118 tests pass (83 unit, 35 integration), locally and on GitHub Actions, in about 5 seconds, with 93% line coverage. The table counts the 93 tests tied to a case; 25 more unit tests cover smaller helpers. To check that the tests can fail, we broke the code on purpose in 8 ways, such as feeding the fire day to the model or treating missing rain as zero, and **all 8 were caught**. **What this does not show.** Passing tests mean the software behaves as specified, not that the forecast is accurate. On real data the model reaches recall 0.949 (Mediterranean test years) and 0.936 (Greece, May-October), but in Greece its AUPRC is below a simple baseline (0.595 vs 0.698, on 47 fires), so SRS criterion AI3(b) is not yet met there. The live FIRMS service and live forecast weather are untested, and the user-acceptance targets still need real users.